frontiers

Supplementary Material

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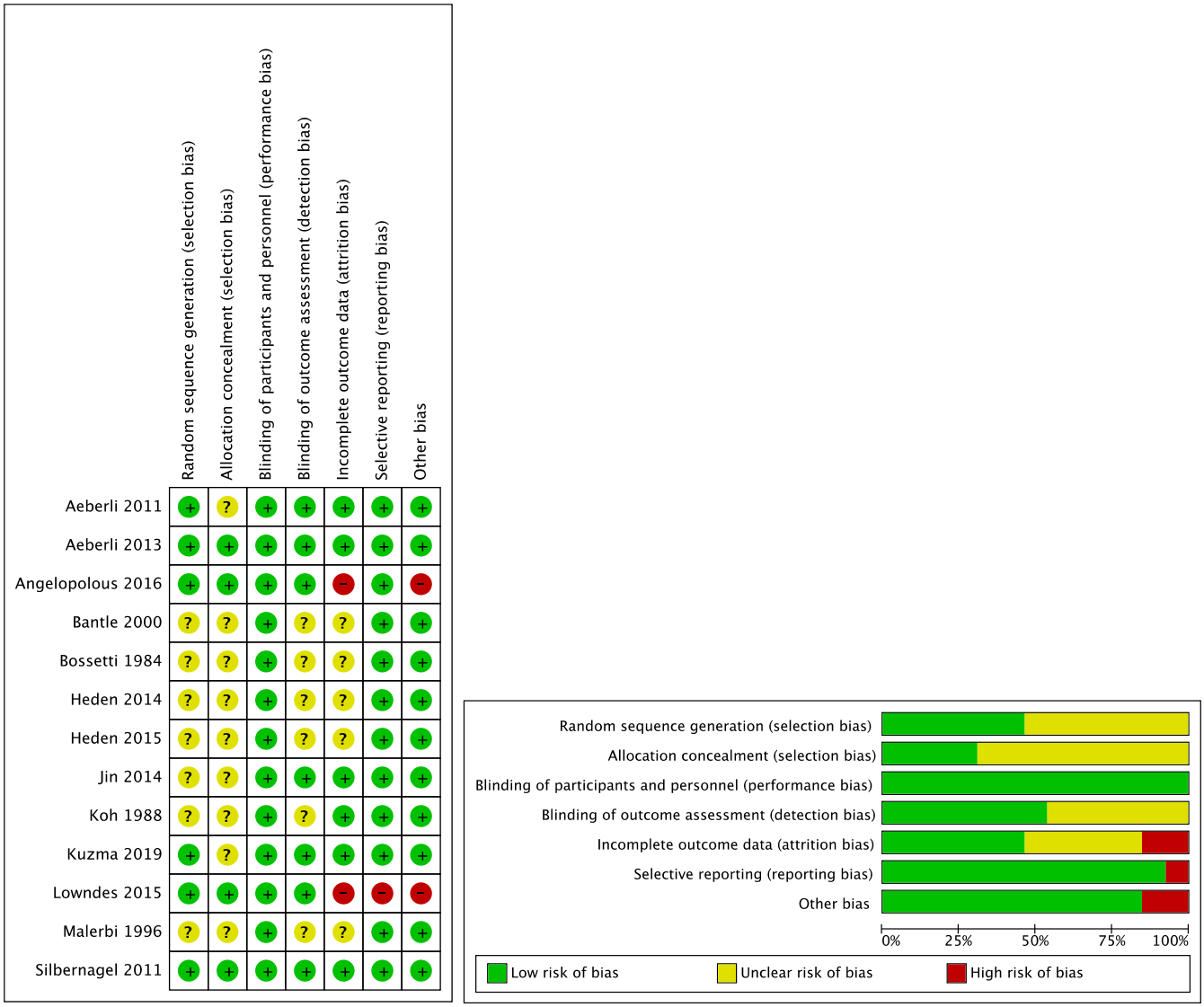
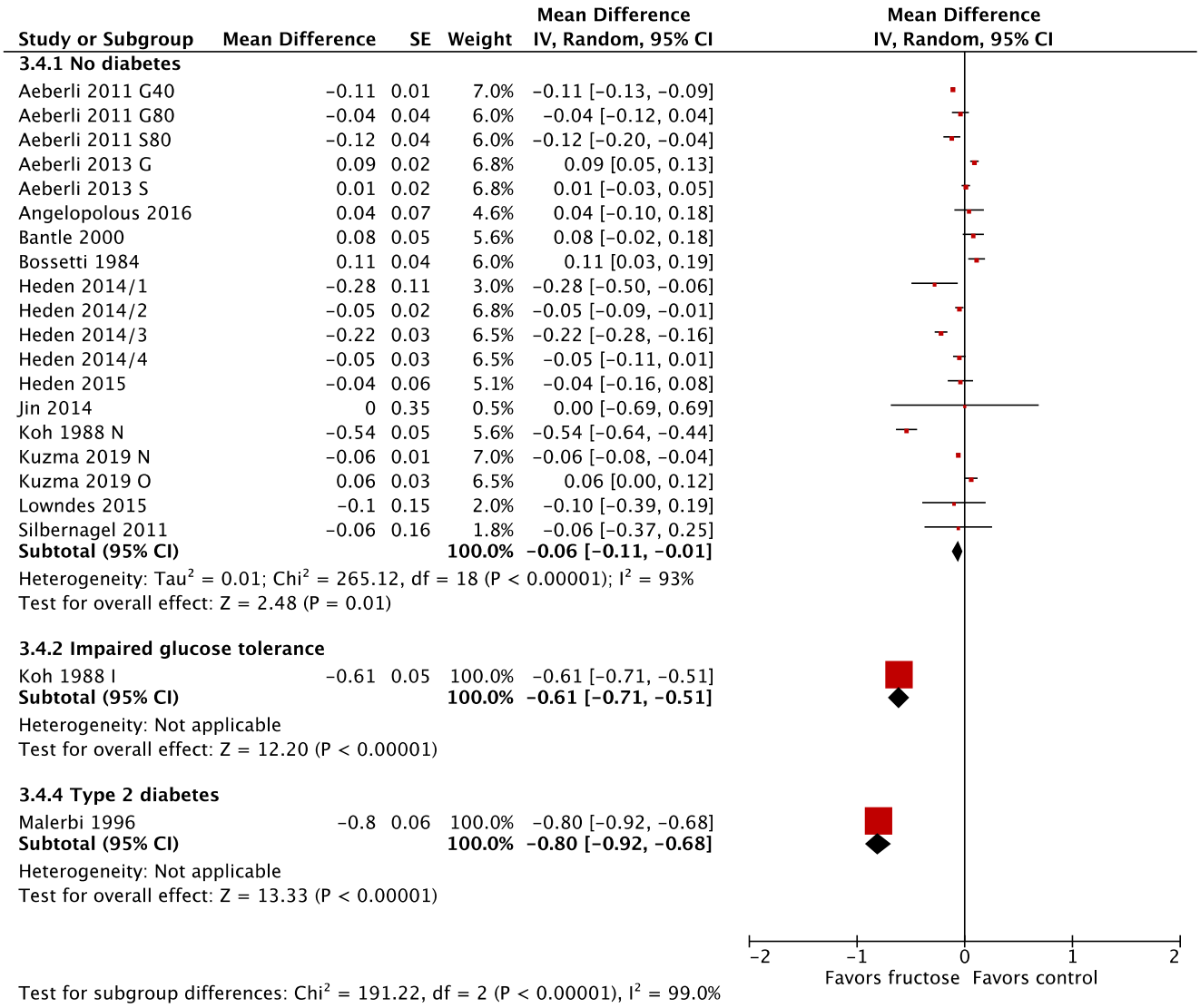
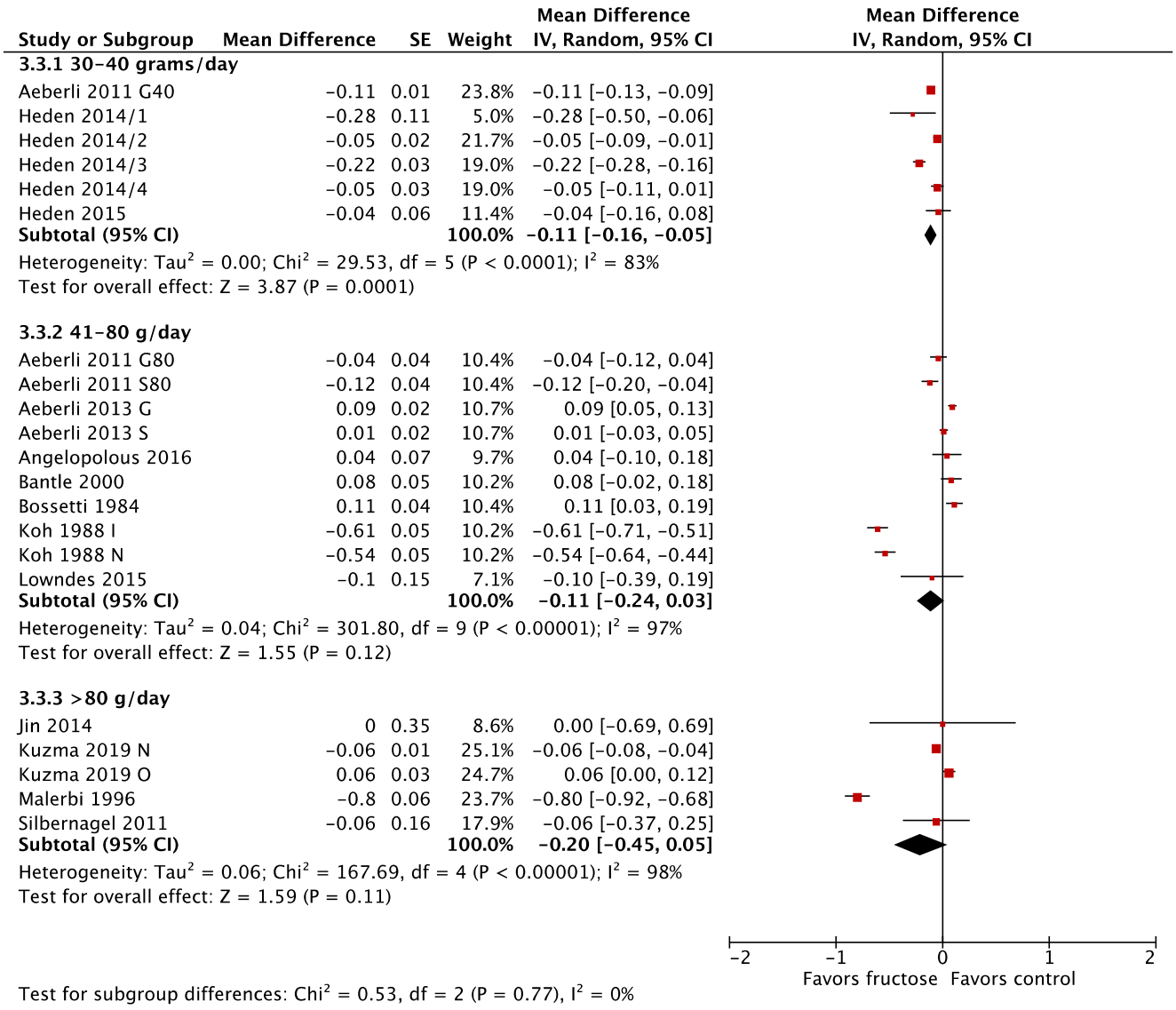


Supplementary figure 1: Risk of bias in included studies.



Supplementary figure 2: Subgroup meta-analysis of fasting blood glucose following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by diabetes status. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood glucose after fructose consumption and fasting blood glucose following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



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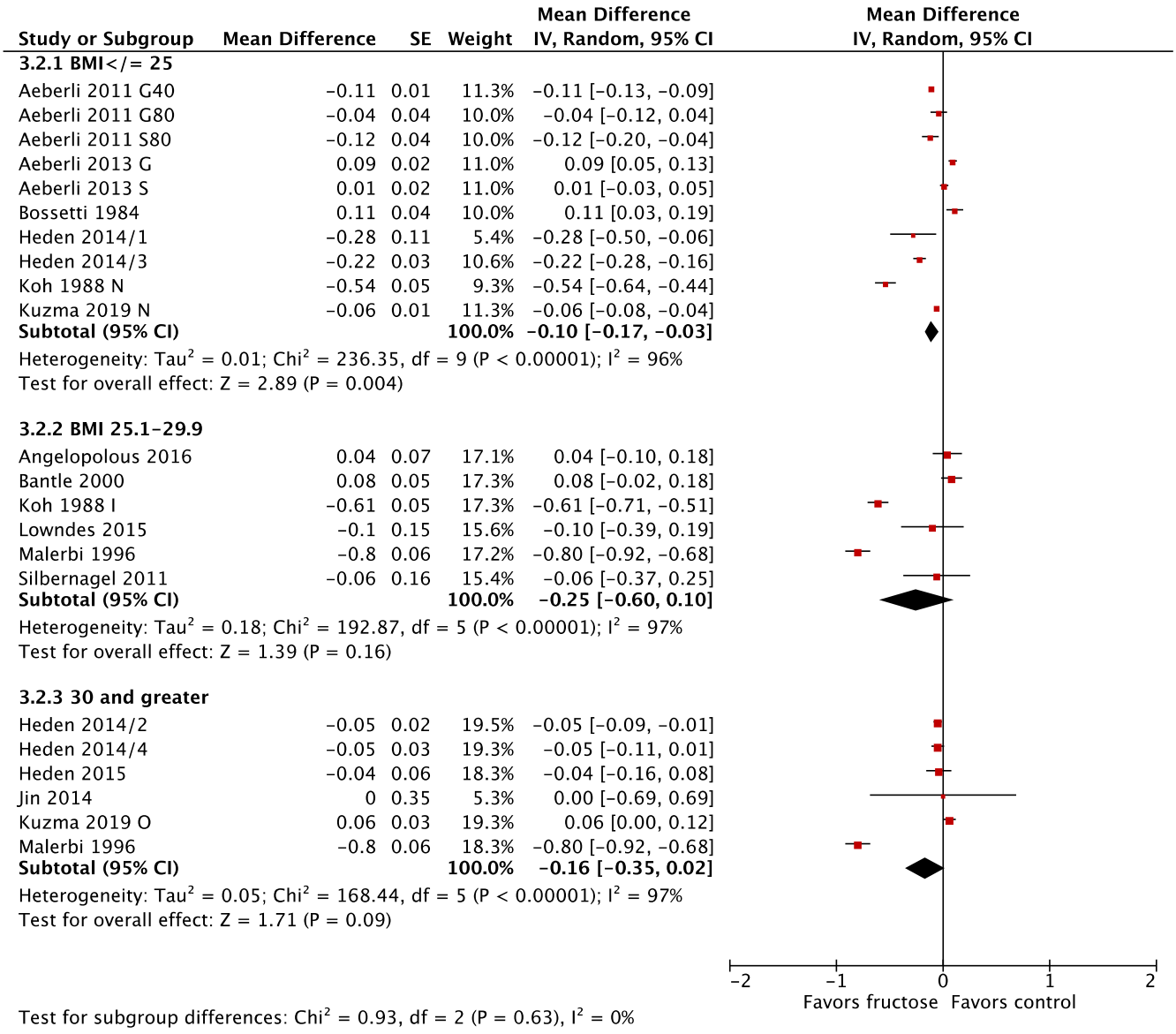
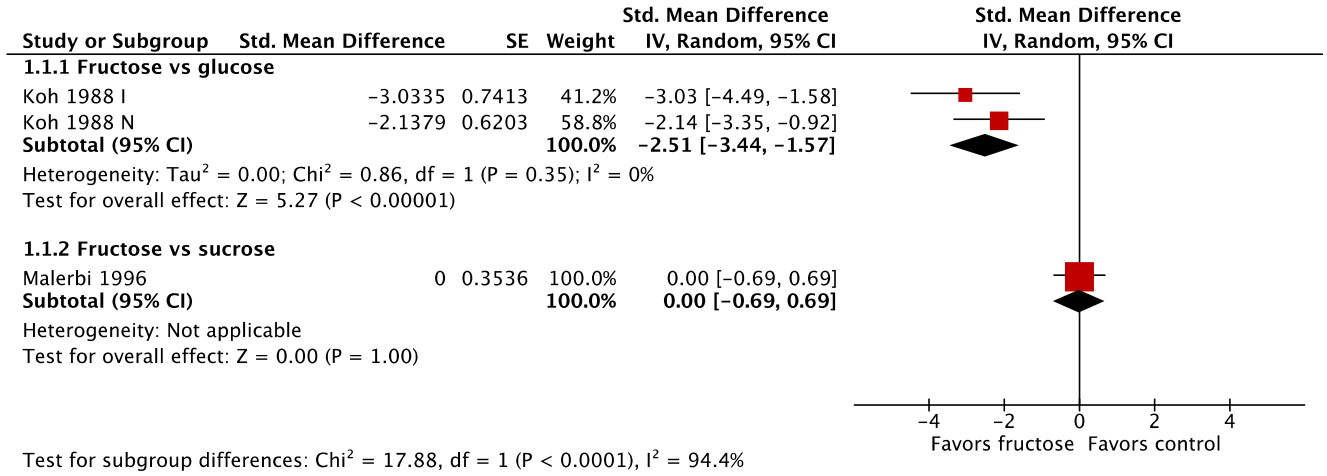
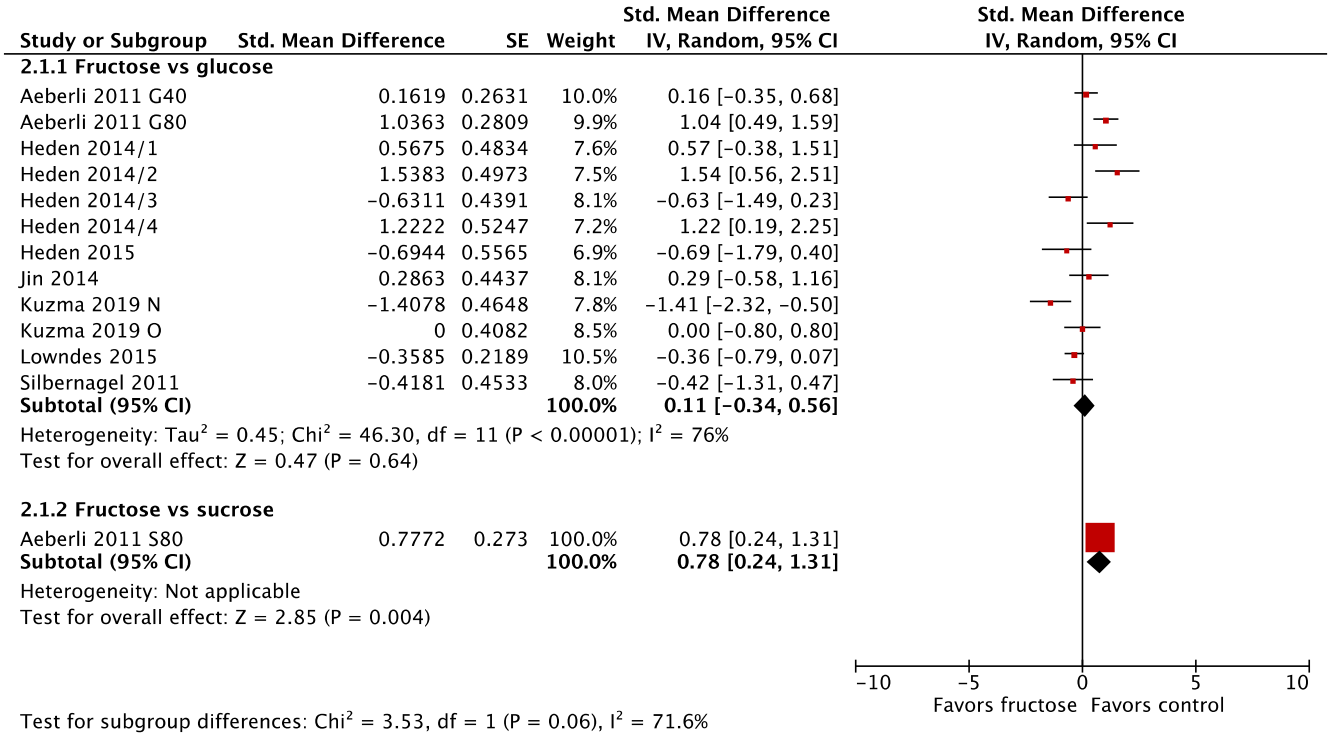


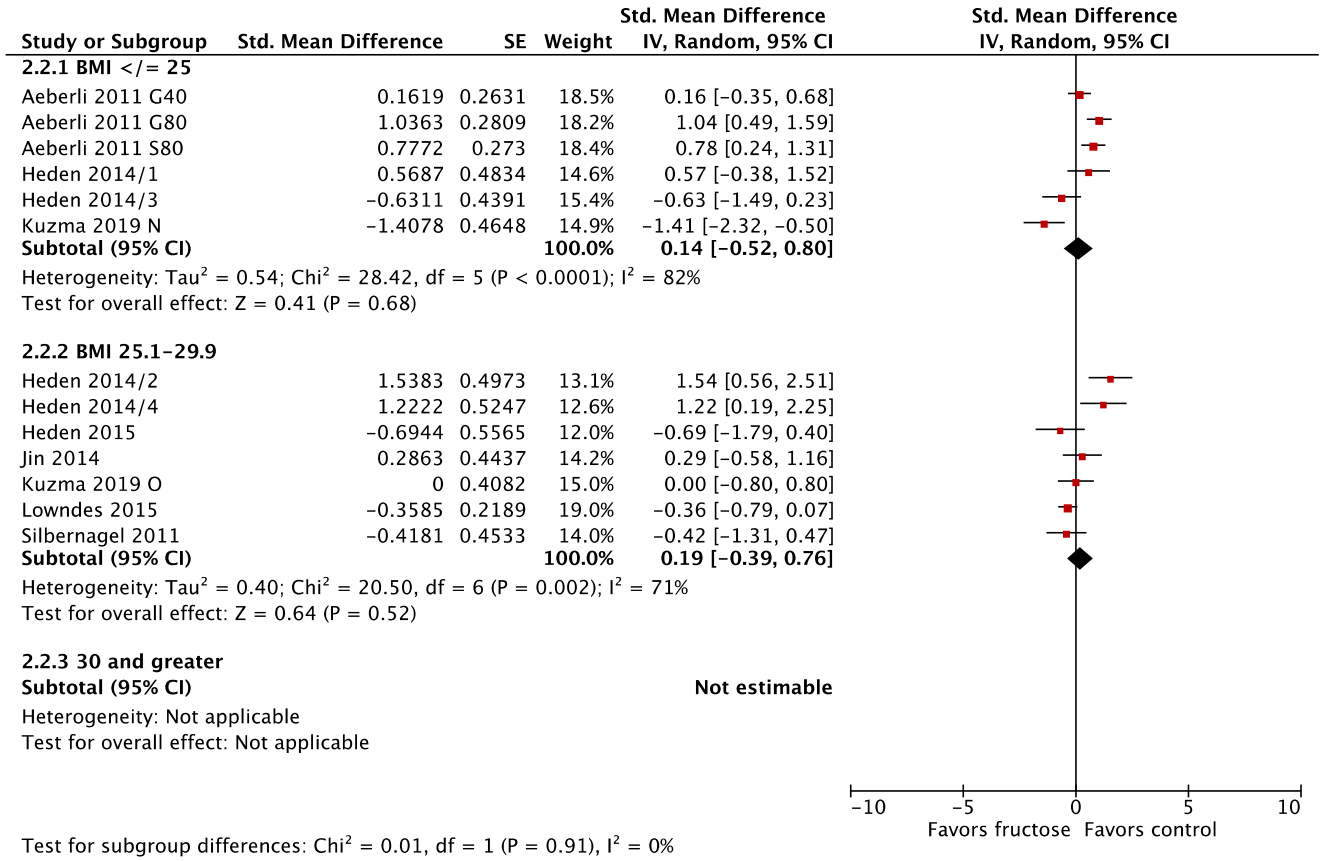
Supplementary figure 4: Subgroup meta-analysis of fasting blood glucose following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by baseline BMI. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood glucose after fructose consumption and fasting blood glucose following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



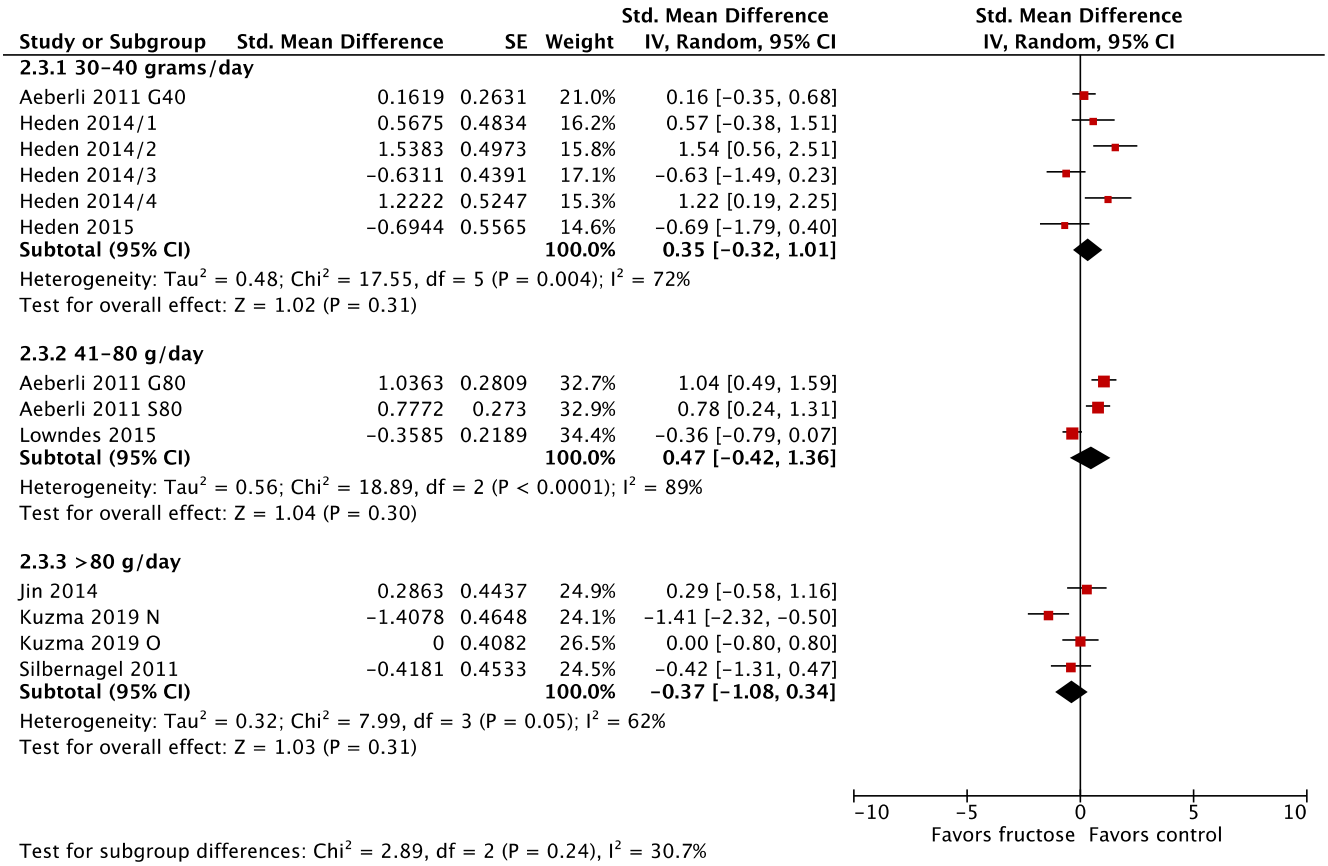
Supplementary figure 5: Subgroup meta-analysis of HbA1c following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by substituted sugar. Values are standardized mean differences [95% CIs] (expressed as units) between HbA1c after fructose consumption and HbA1c following glucose or sucrose consumption. IV, inverse variance; SE, standard error; I, impaired glucose tolerance, N, body weight



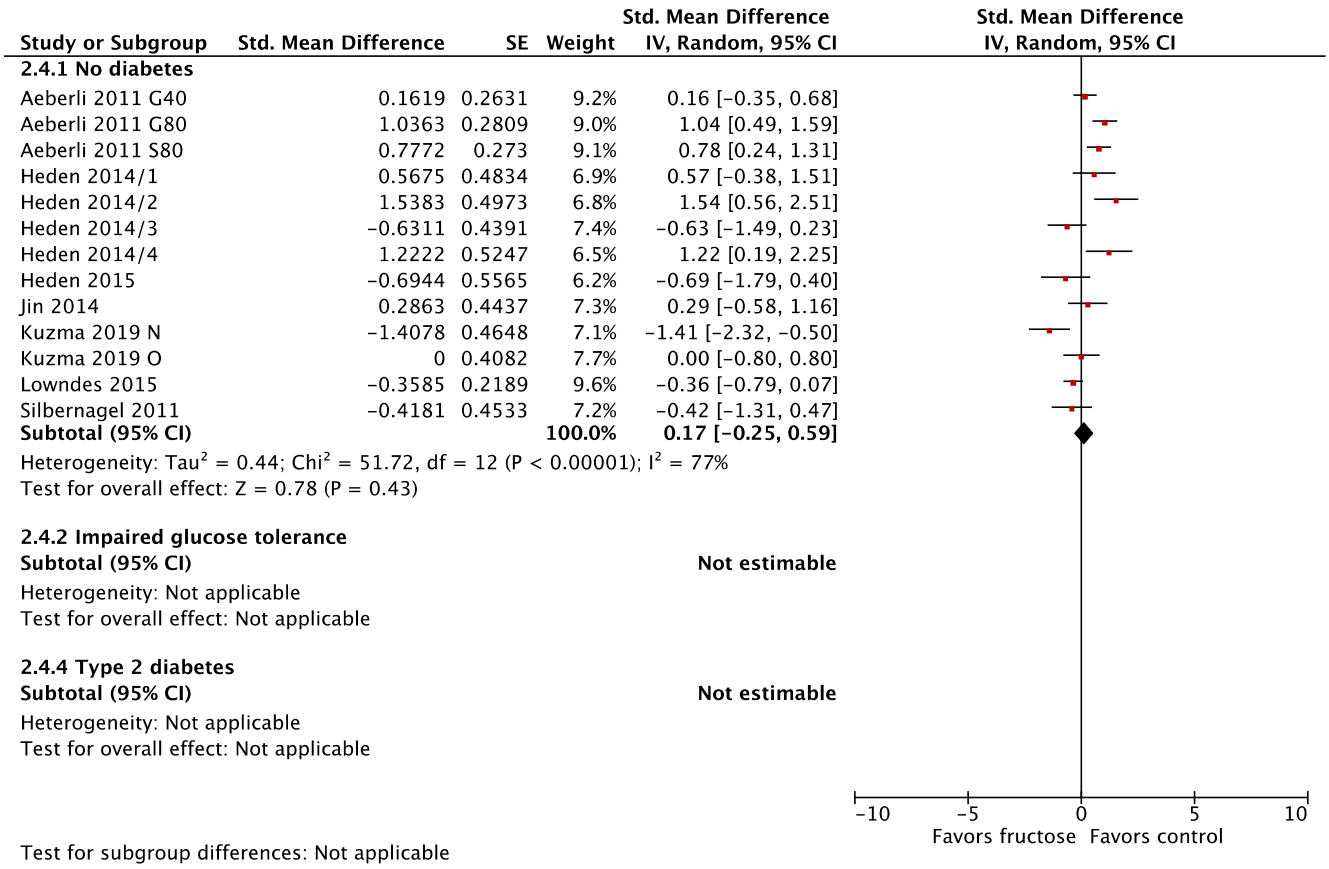
Supplementary figure 6: Subgroup meta-analysis of HOMA-IR/HOMA2 following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by substituted sugar. Values are standardized mean differences [95% CIs] (expressed as units) between HOMA-IR or HOMA2 after fructose consumption and HOMA-IR or HOMA2 following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; N, normal body weight; O, overweight



Supplementary figure 7: Subgroup meta-analysis of HOMA-IR/HOMA2 following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by baseline BMI. Values are standardized mean differences [95% CIs] (expressed as units) between HOMA-IR or HOMA2 after fructose consumption and HOMA-IR or HOMA2 following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; N, normal body weight; O, overweight



Supplementary figure 8: Subgroup meta-analysis of HOMA-IR/HOMA2 following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by dose of sugar. Values are standardized mean differences [95% CIs] (expressed as units) between HOMA-IR or HOMA2 after fructose consumption and HOMA-IR or HOMA2 following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; N, normal body weight; O, overweight



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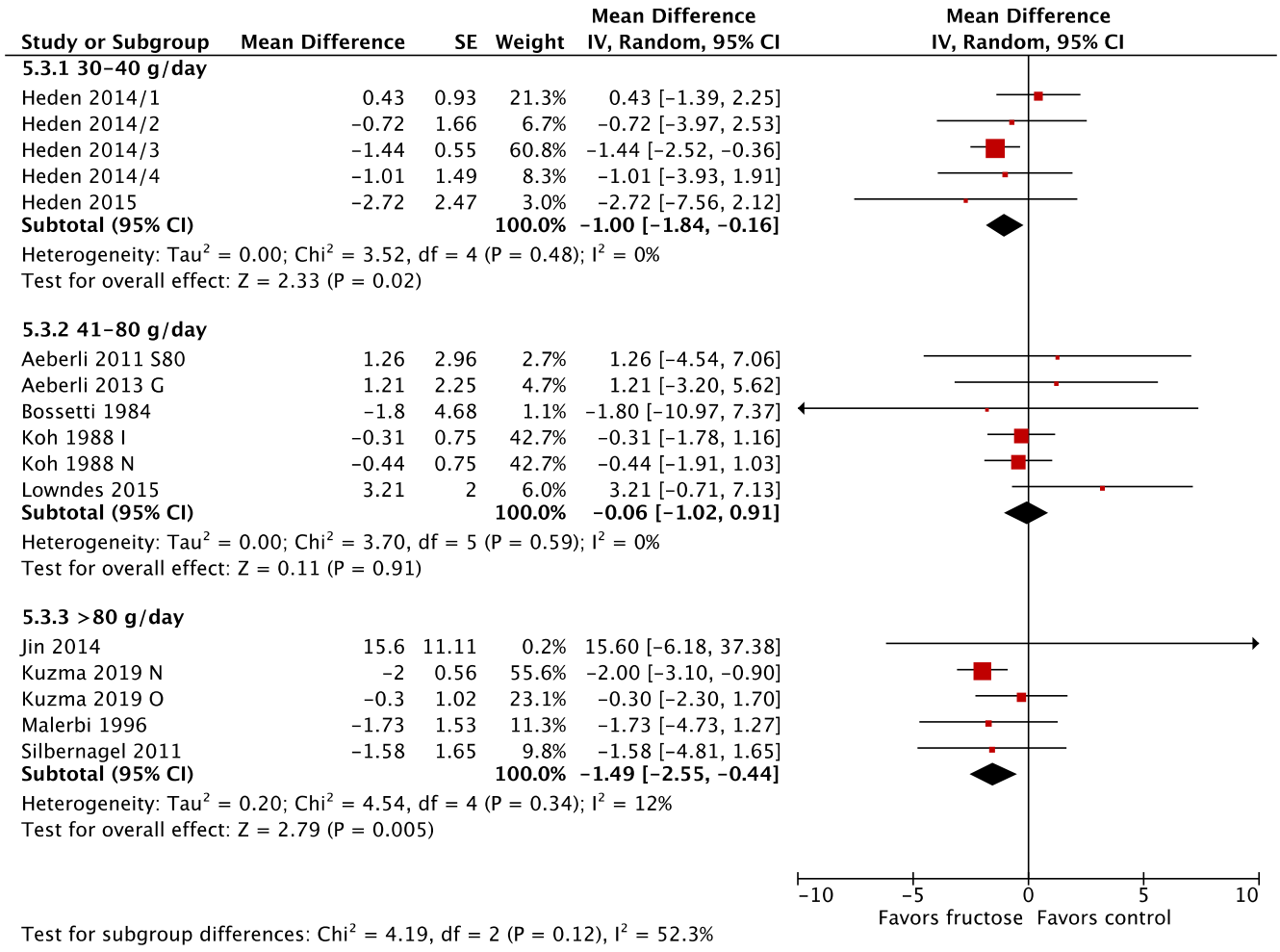
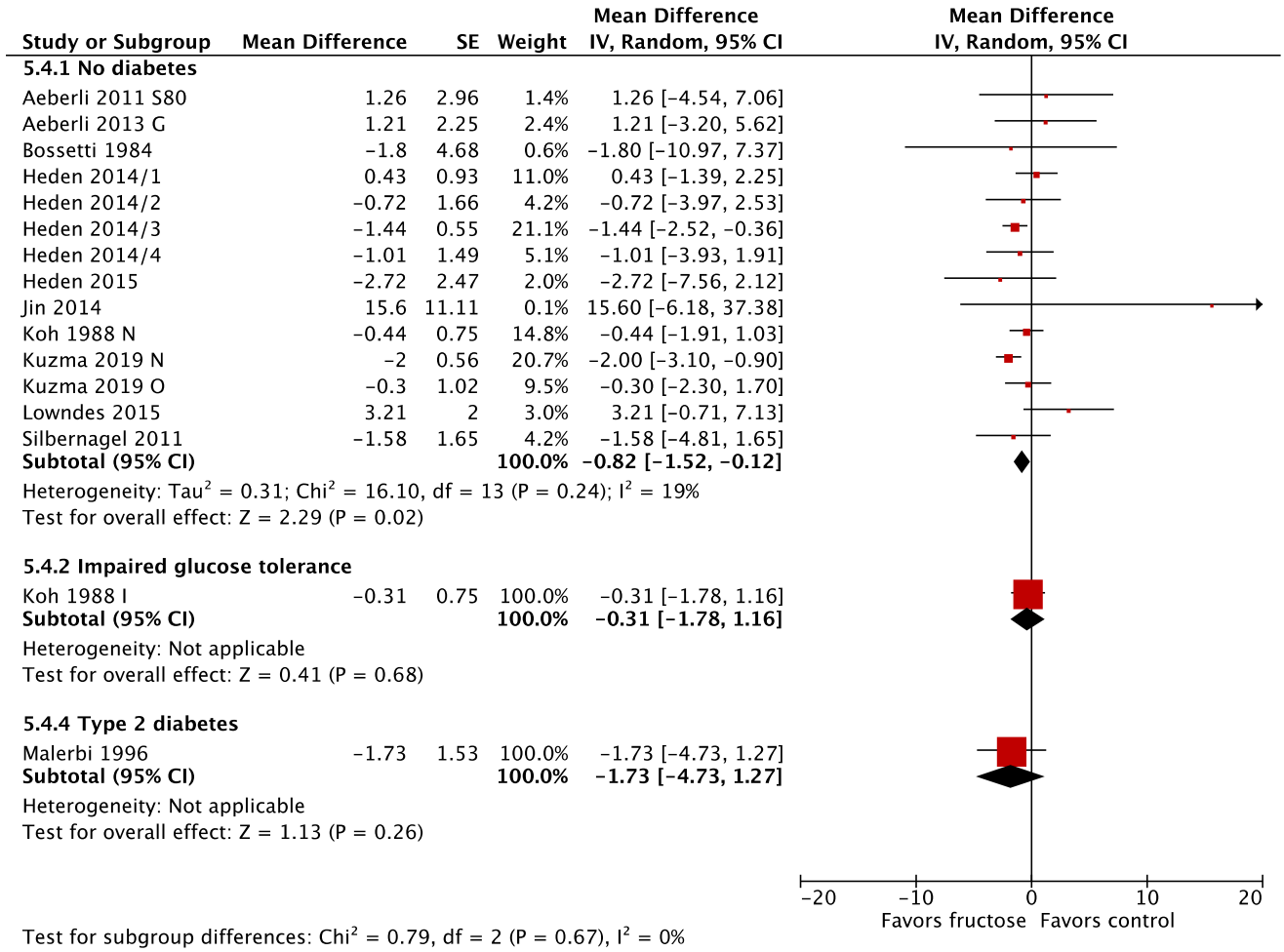


Supplementary figure 10: Subgroup meta-analysis of fasting blood insulin following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by dose of sugar. Values are mean differences [95% CIs] (expressed as µIU/mL) between fasting blood insulin after fructose consumption and fasting blood insulin following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



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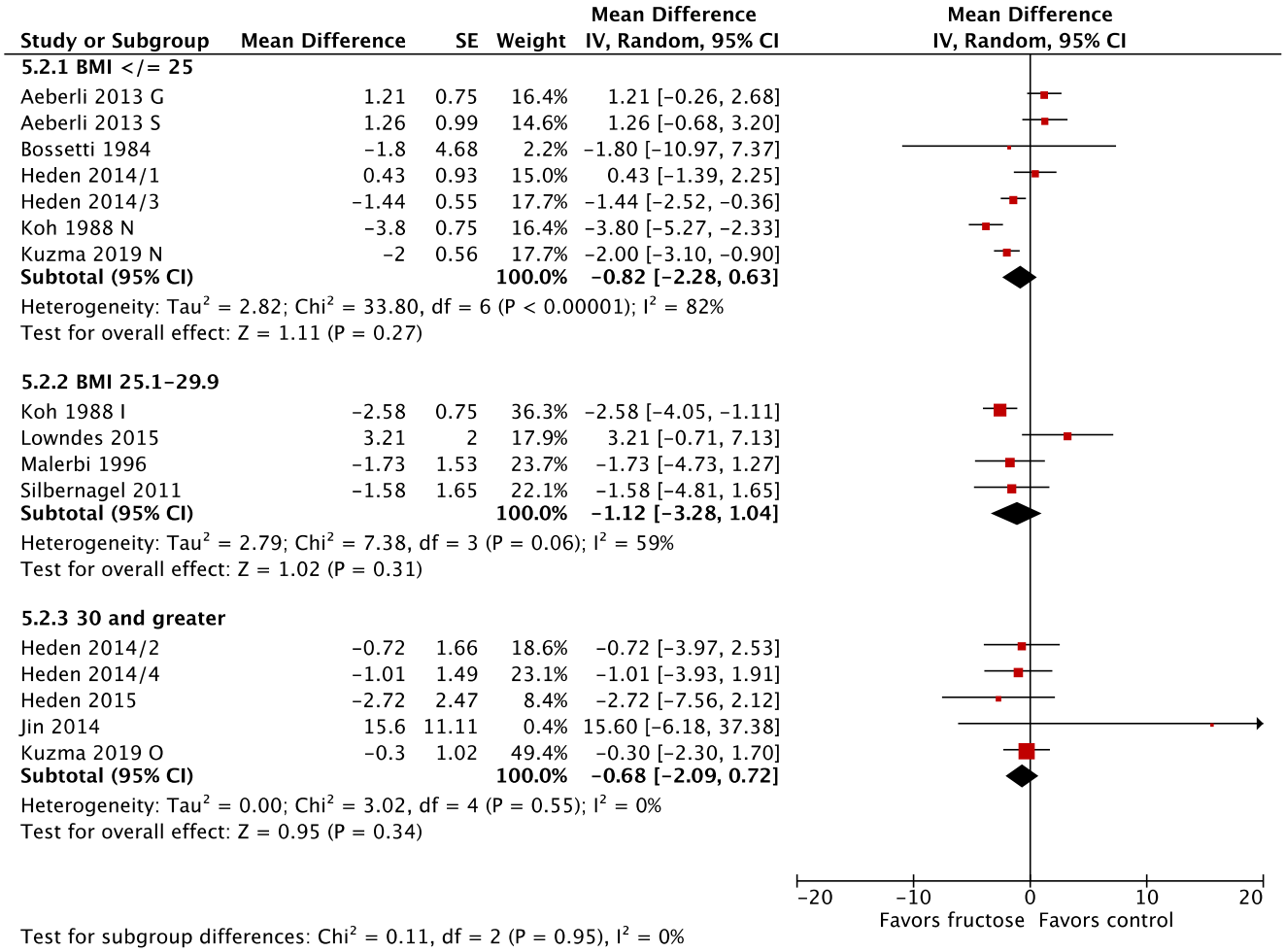
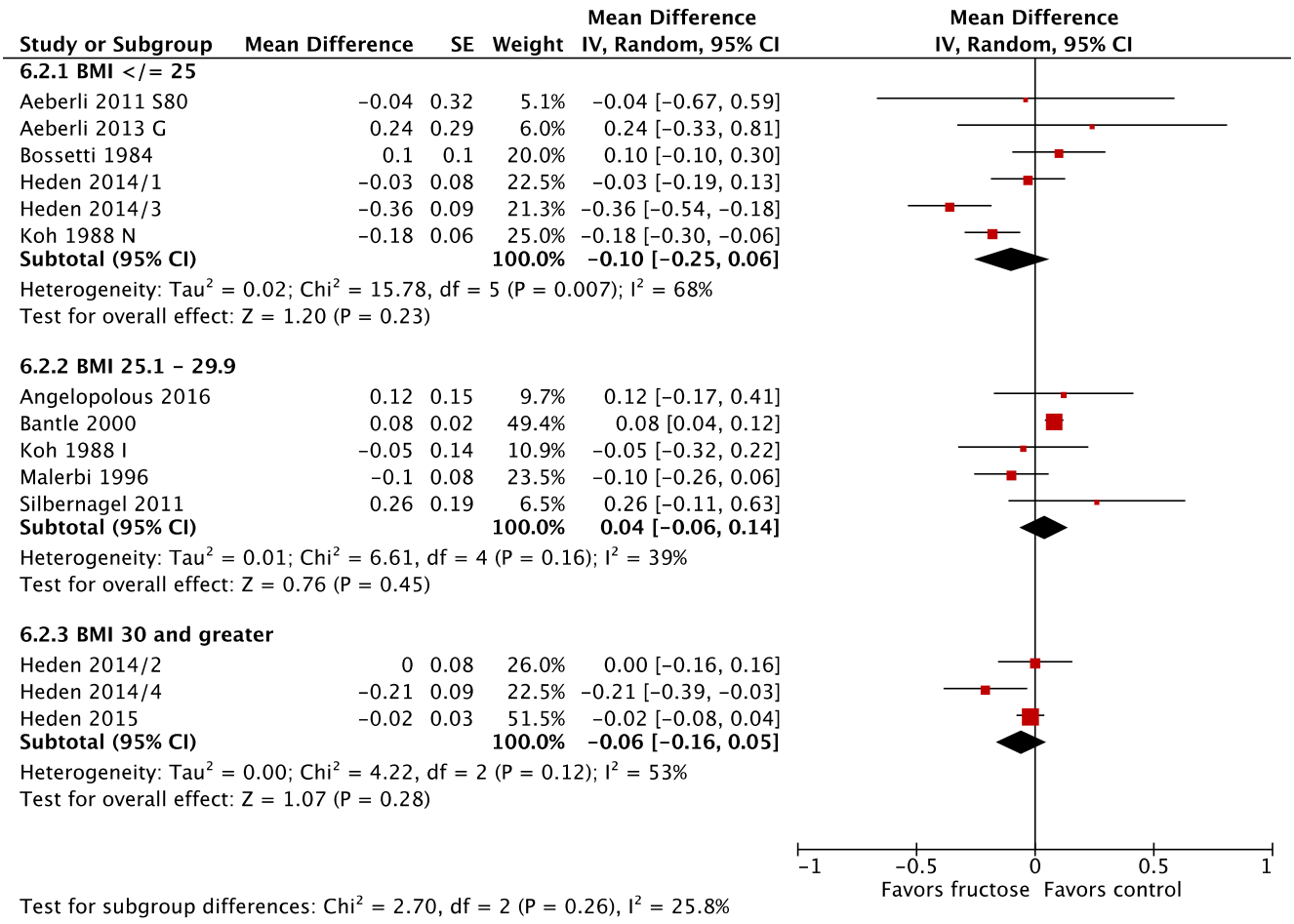
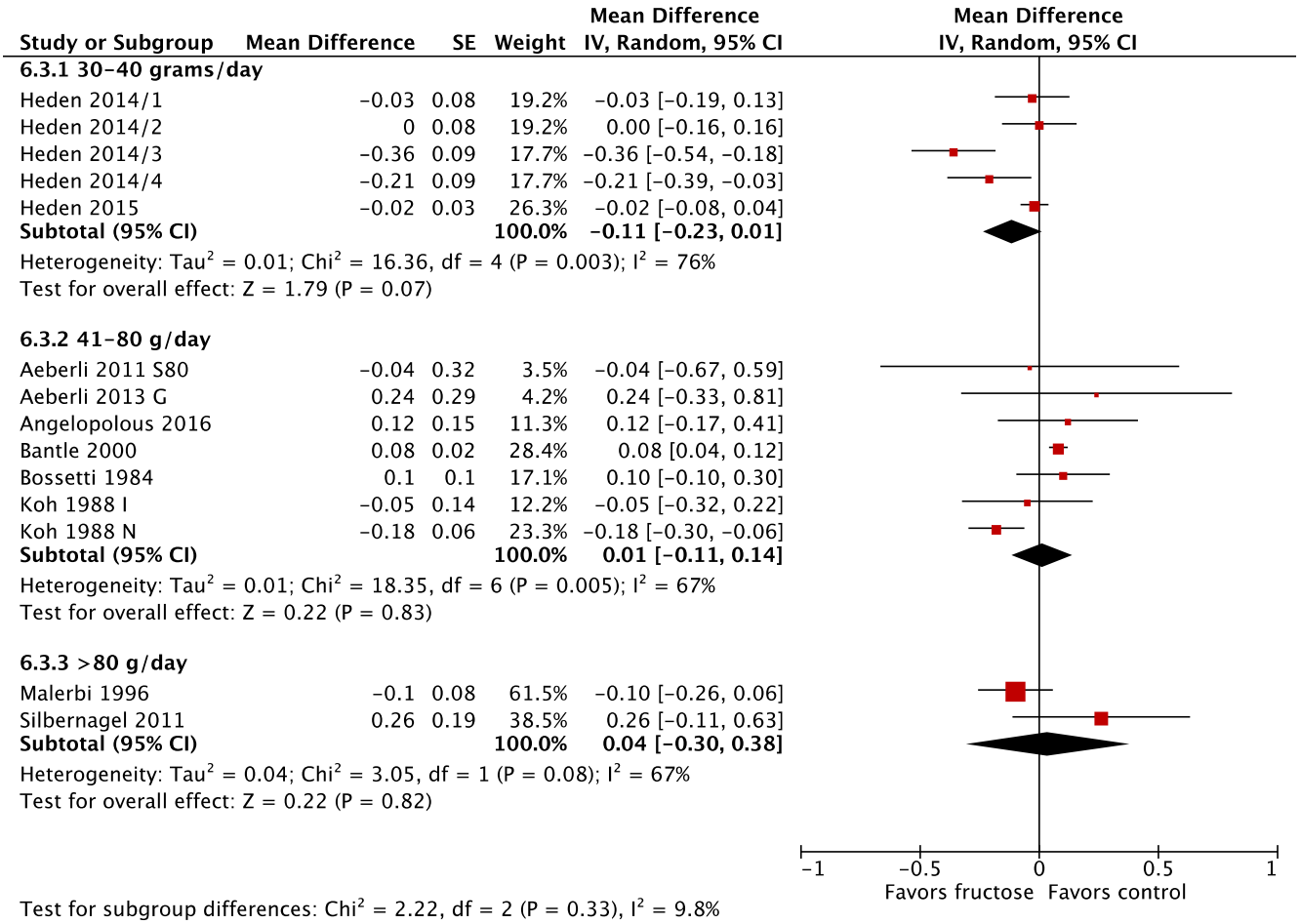


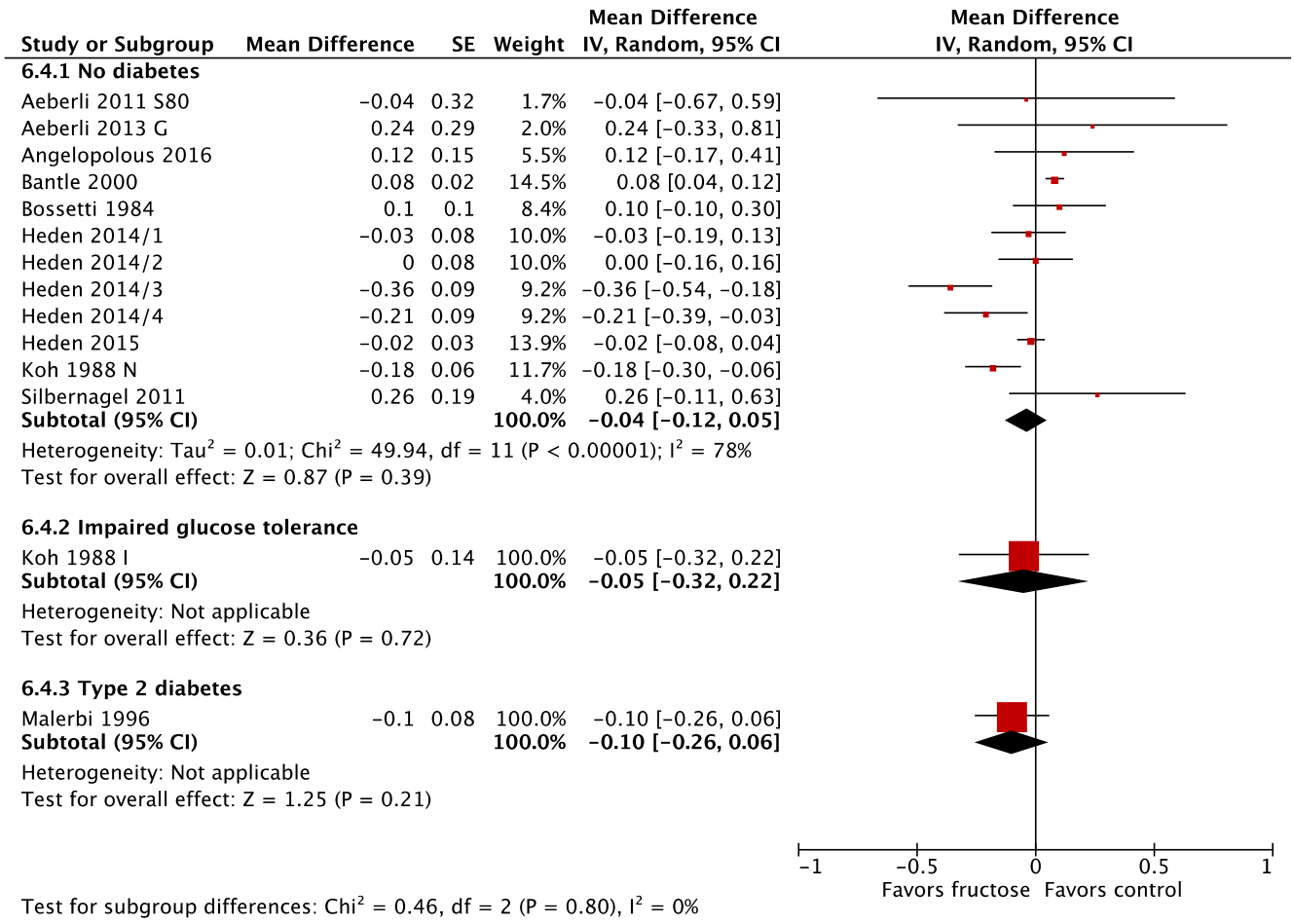
Supplementary figure 12: Subgroup meta-analysis of fasting blood insulin following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by baseline BMI. Values are mean differences [95% CIs] (expressed as µIU/mL) between fasting blood insulin after fructose consumption and fasting blood insulin following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



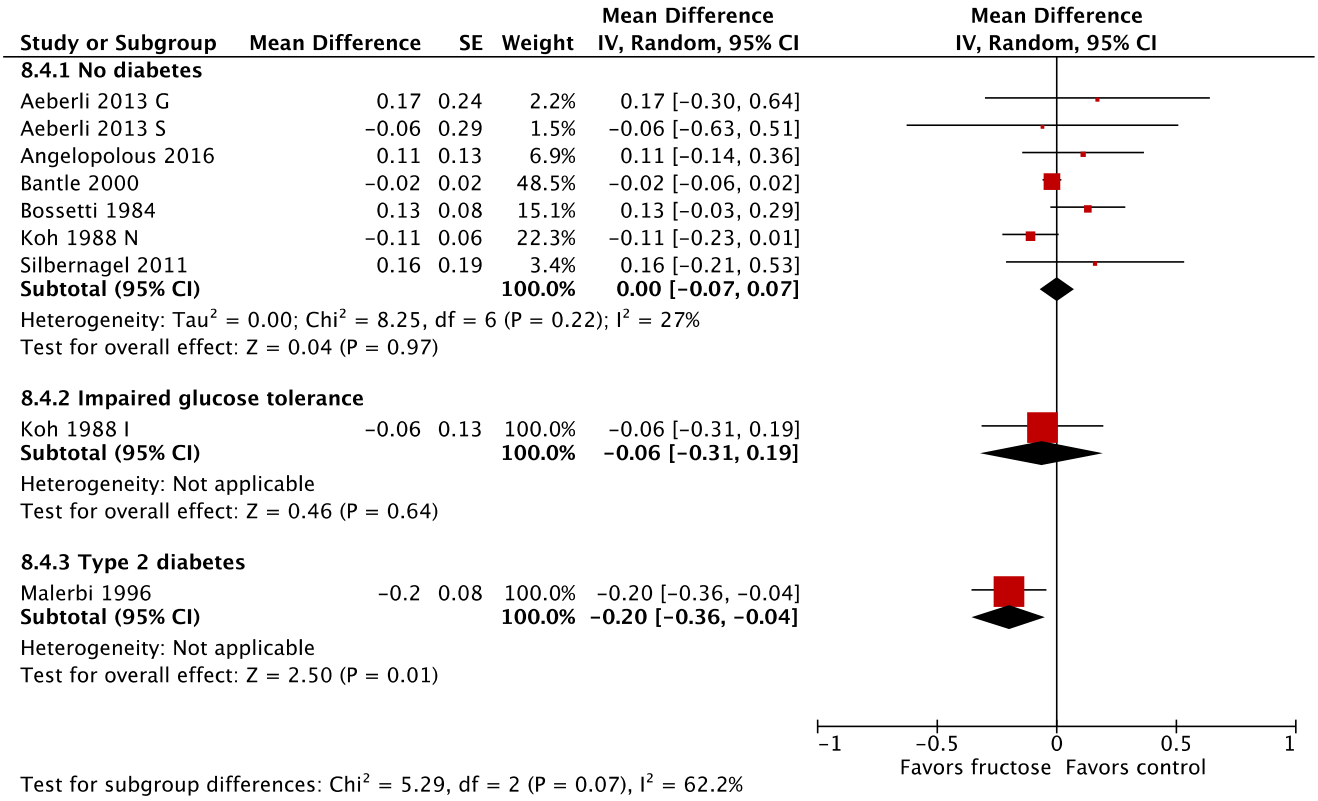
Supplementary figure 13: Subgroup meta-analysis of fasting blood total cholesterol following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by baseline BMI. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood total cholesterol after fructose consumption and fasting blood total cholesterol following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



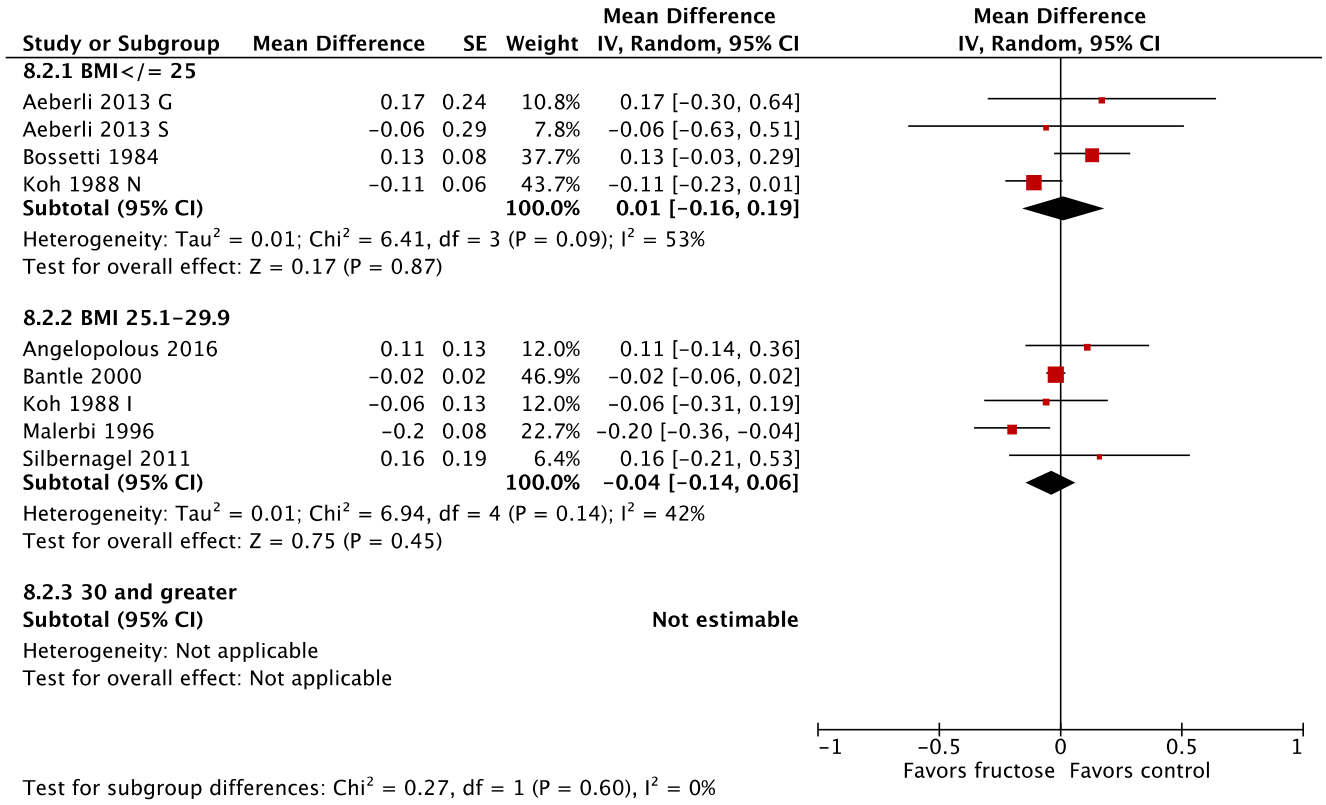
Supplementary figure 14: Subgroup meta-analysis of fasting blood total cholesterol following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by dose of sugar. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood total cholesterol after fructose consumption and fasting blood total cholesterol following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



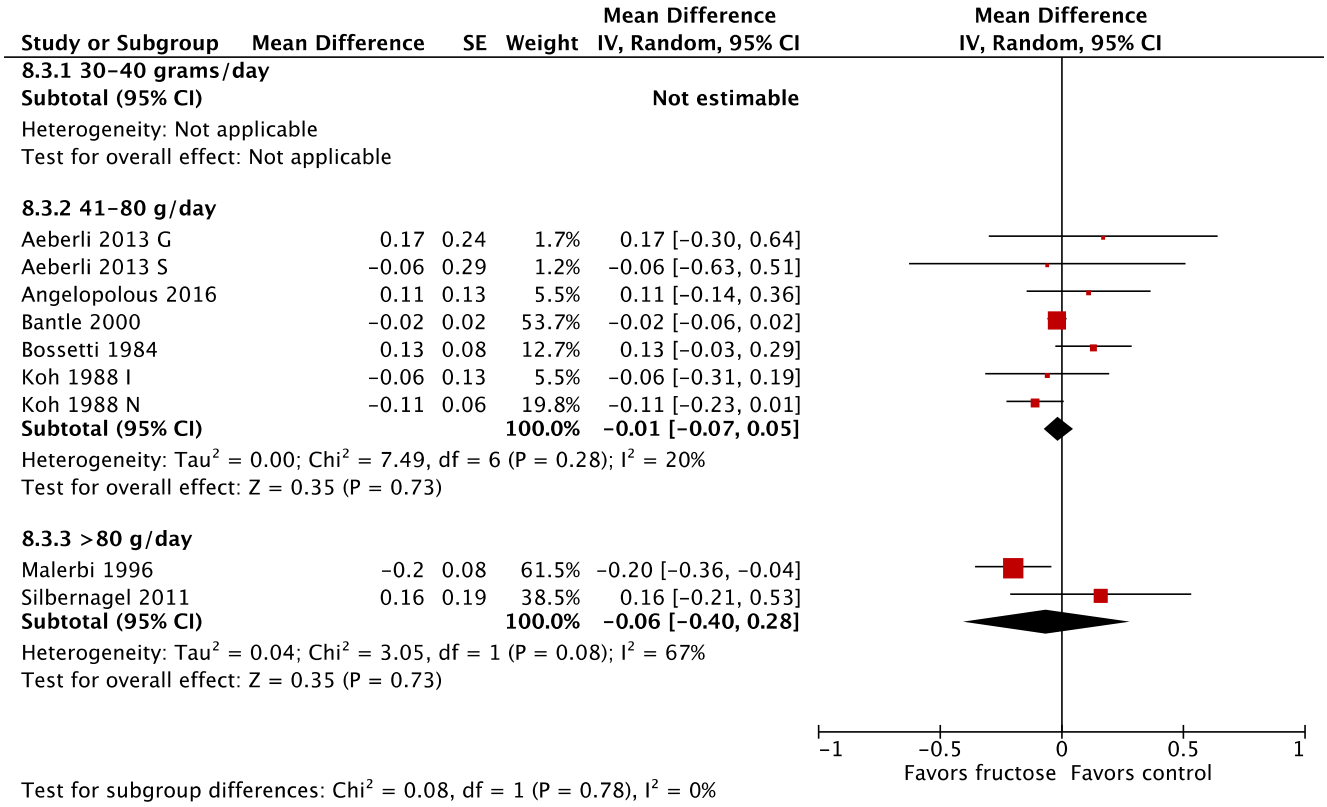
Supplementary figure 15: Subgroup meta-analysis of fasting blood total cholesterol following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by diabetes status. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood total cholesterol after fructose consumption and fasting blood total cholesterol following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



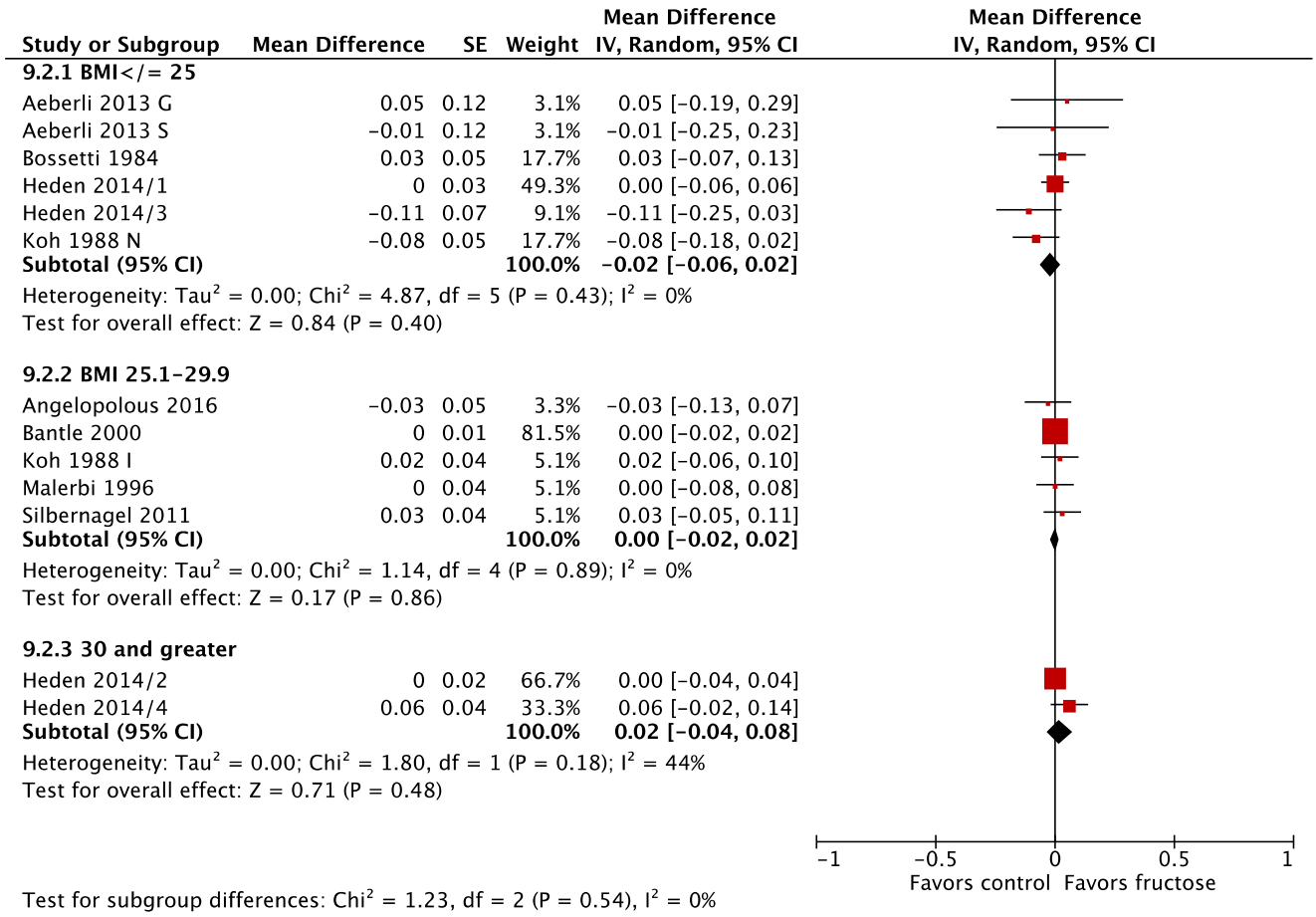
Supplementary figure 16: Subgroup meta-analysis of fasting LDL cholesterol following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by diabetes status. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood LDL cholesterol after fructose consumption and fasting blood LDL cholesterol following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



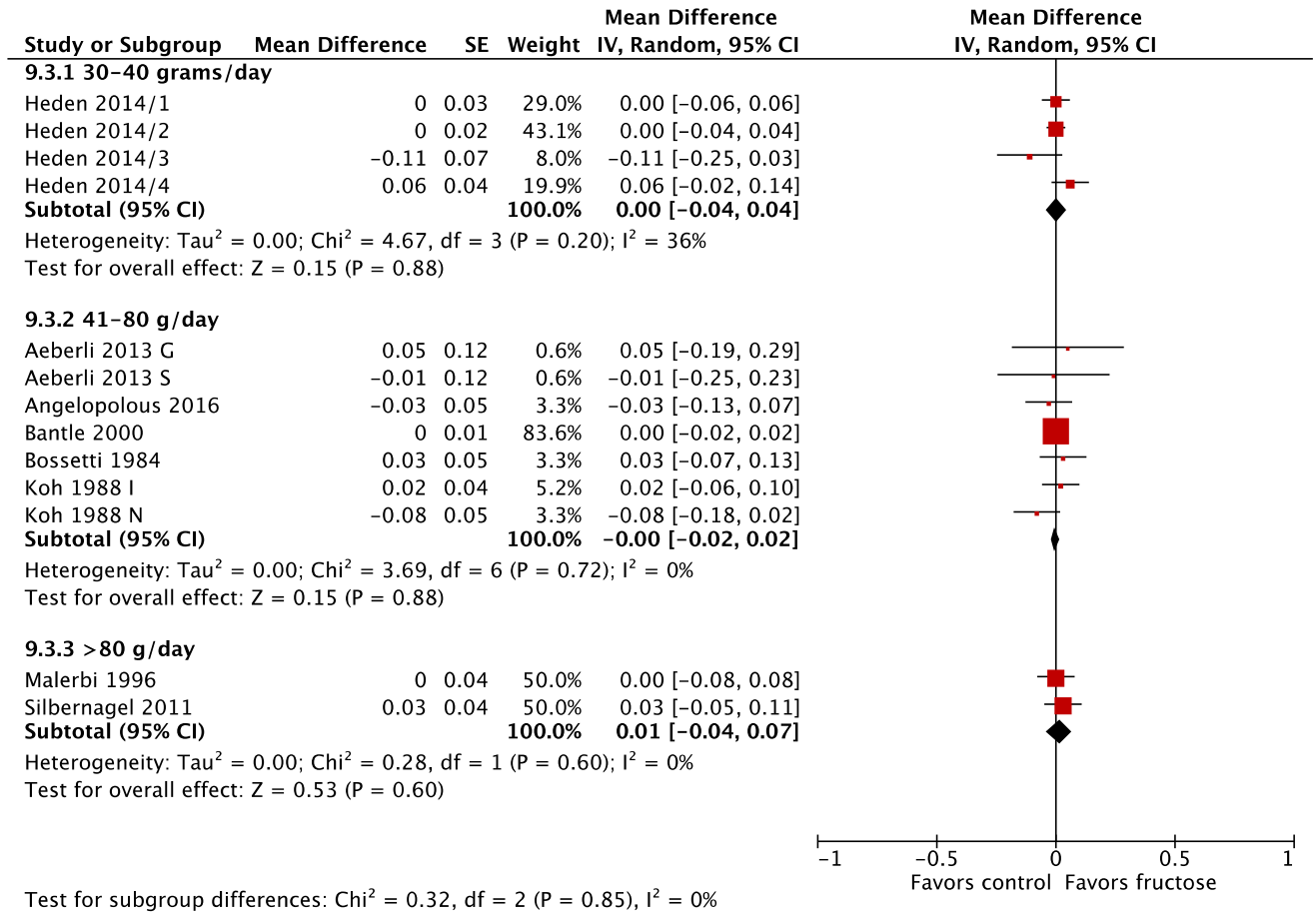
Supplementary figure 17: Subgroup meta-analysis of fasting LDL cholesterol following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by baseline BMI. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood LDL cholesterol after fructose consumption and fasting blood LDL cholesterol following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



Supplementary figure 18: Subgroup meta-analysis of fasting LDL cholesterol following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by dose of sugar. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood LDL cholesterol after fructose consumption and fasting blood LDL cholesterol following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



Supplementary figure 19: Subgroup meta-analysis of fasting HDL cholesterol following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by baseline BMI. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood HDL cholesterol after fructose consumption and fasting blood HDL cholesterol following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



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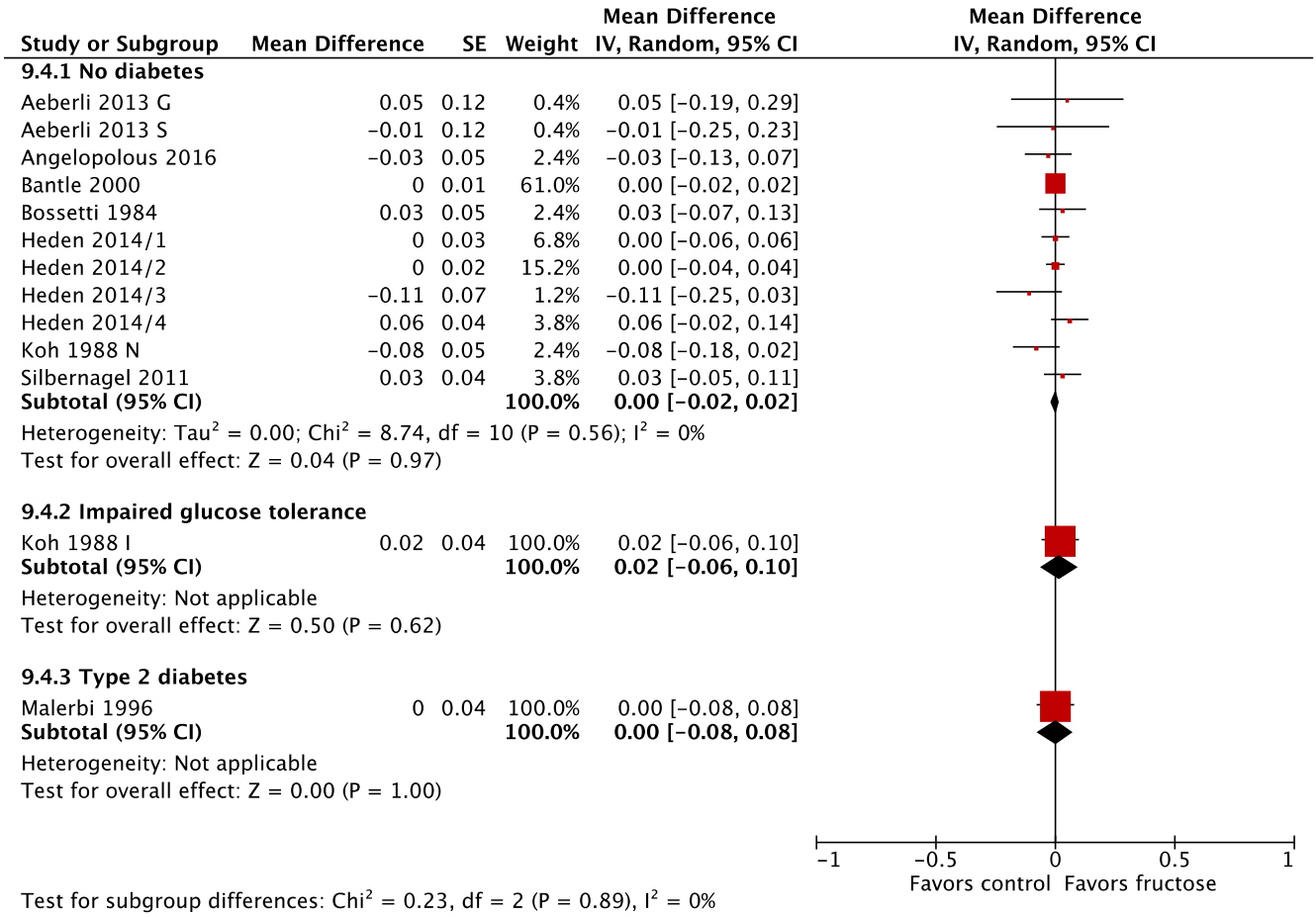
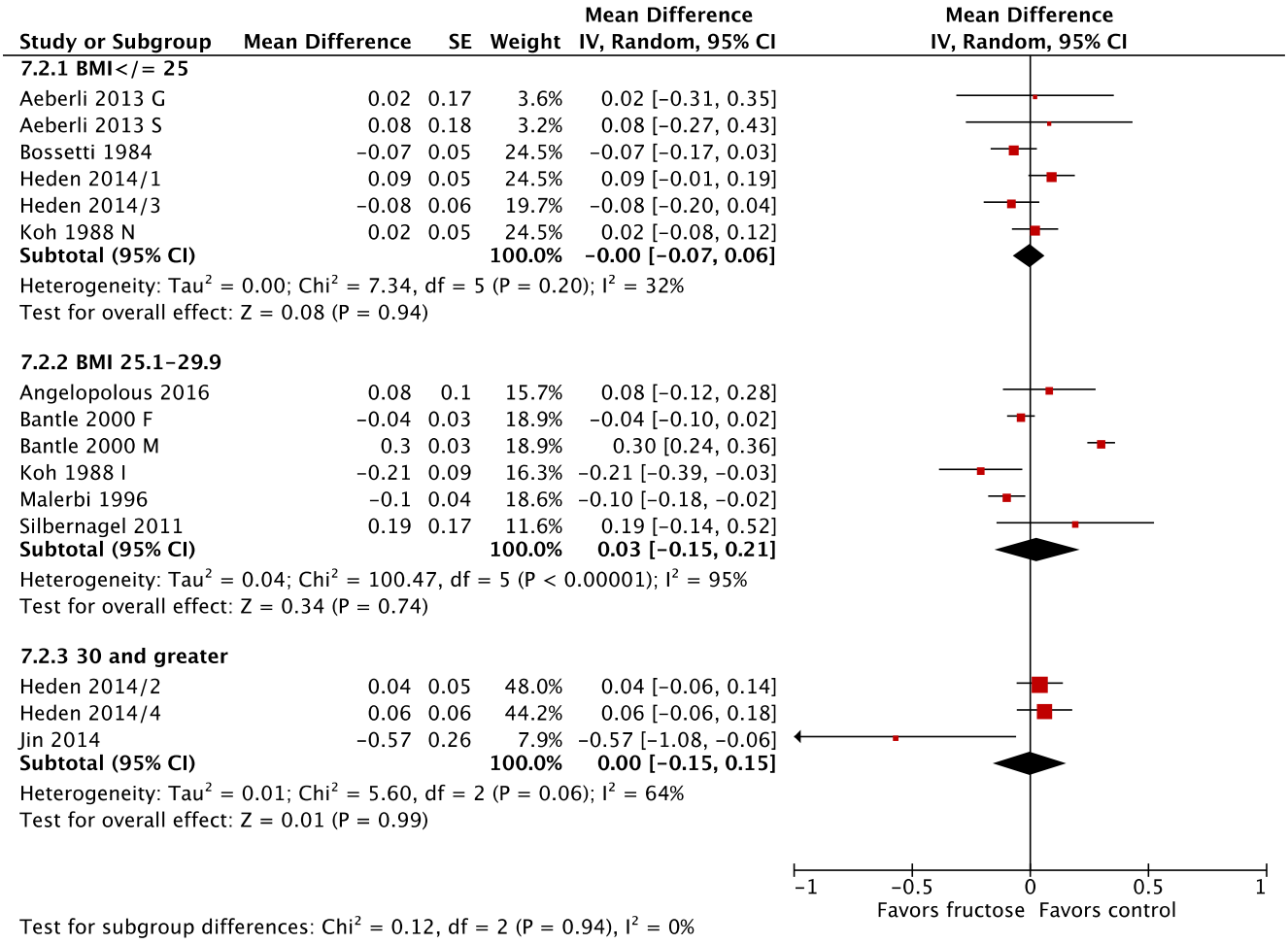
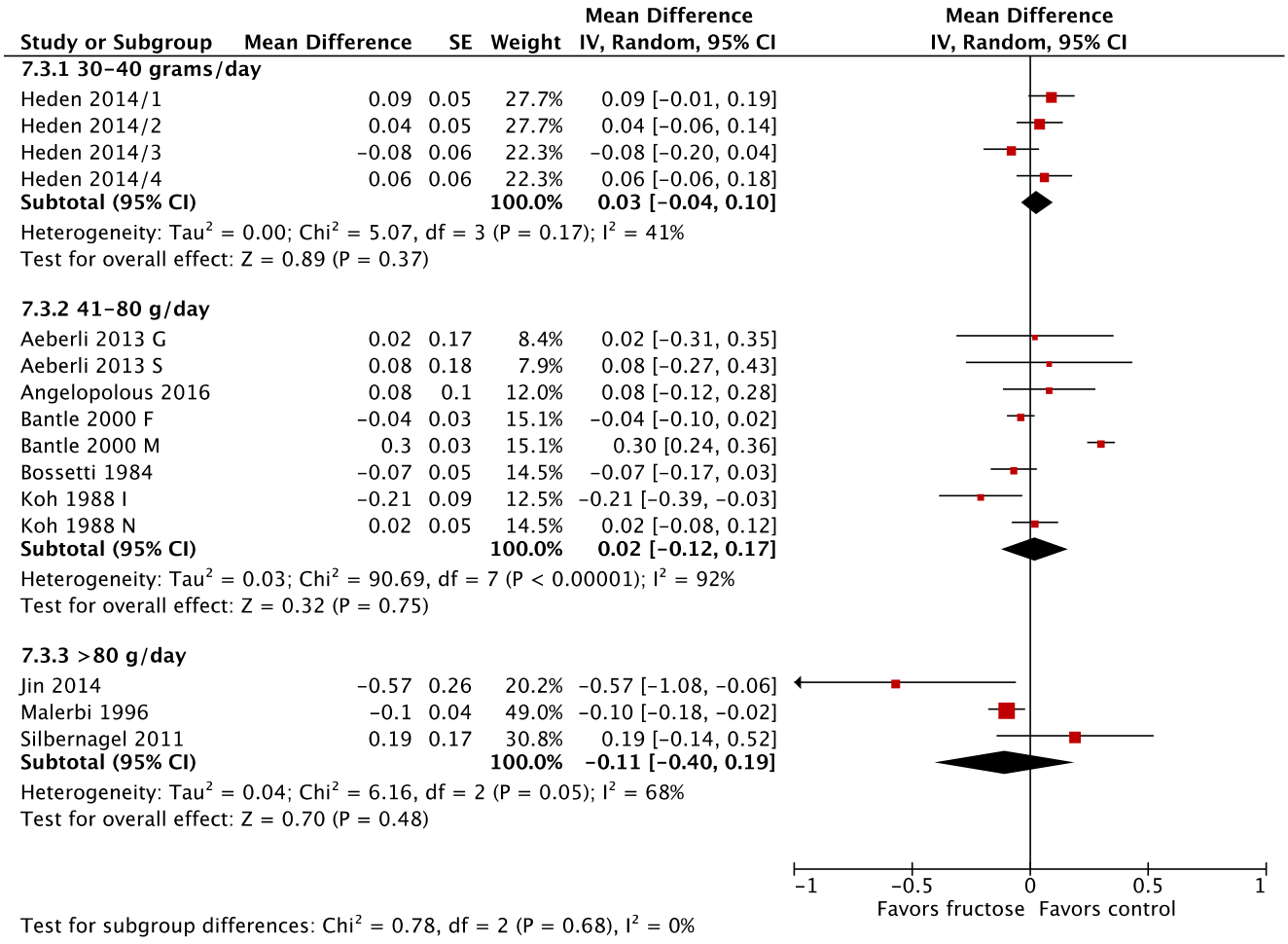


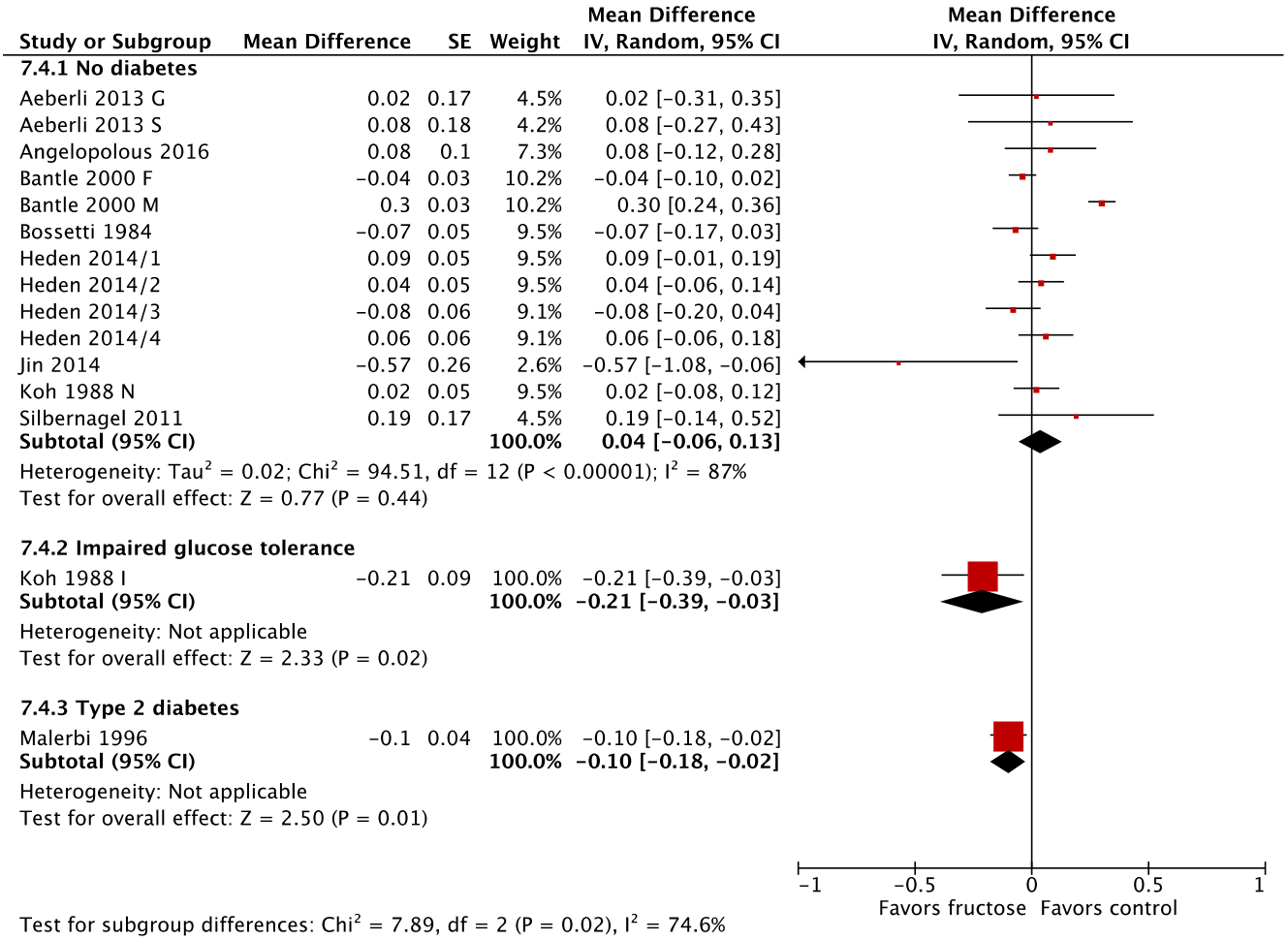
Supplementary figure 21: Subgroup meta-analysis of fasting HDL cholesterol following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by diabetes status. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood HDL cholesterol after fructose consumption and fasting blood HDL cholesterol following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



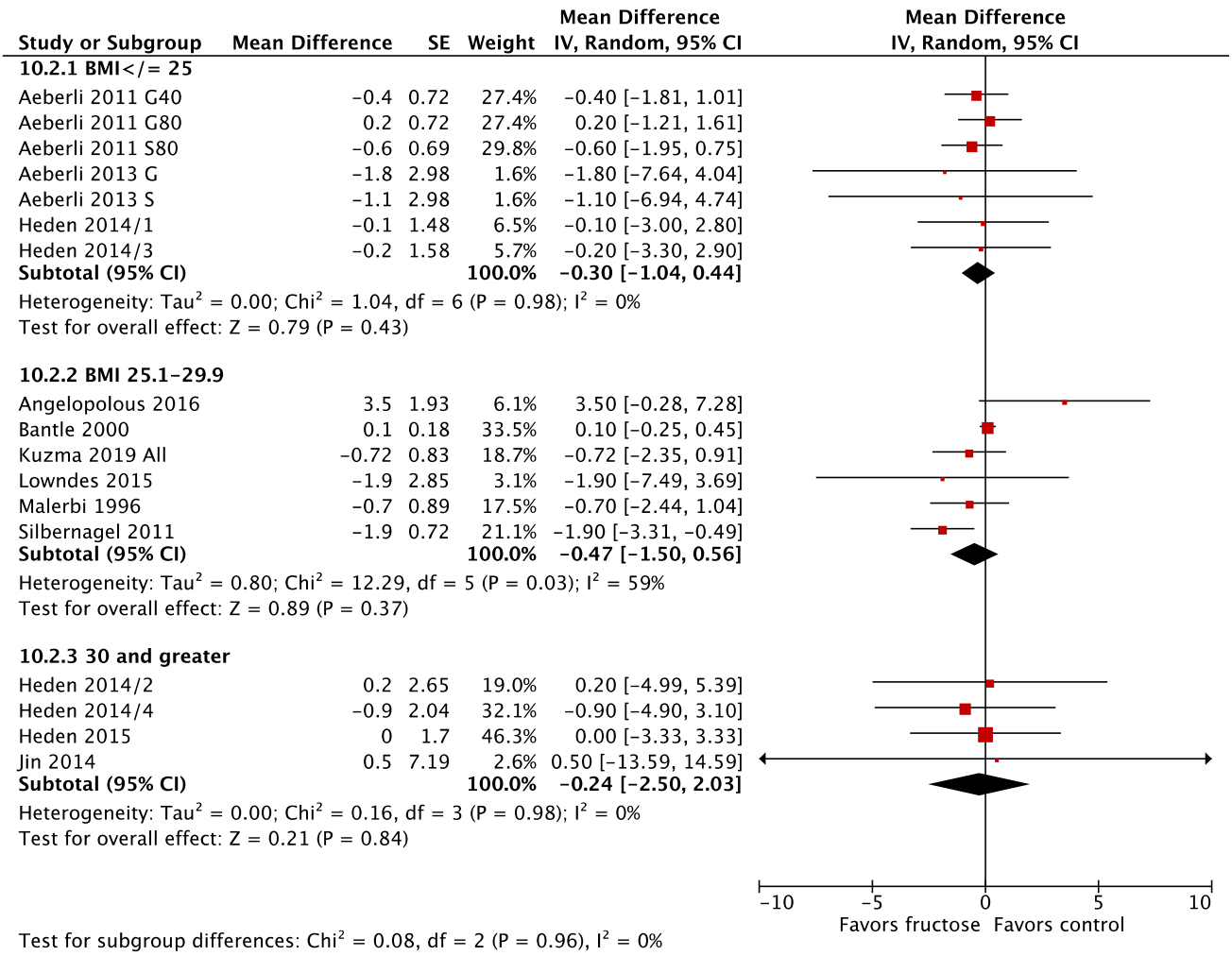
Supplementary figure 22: Subgroup meta-analysis of fasting blood triglycerides following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by baseline BMI. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood triglycerides after fructose consumption and fasting blood triglycerides following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



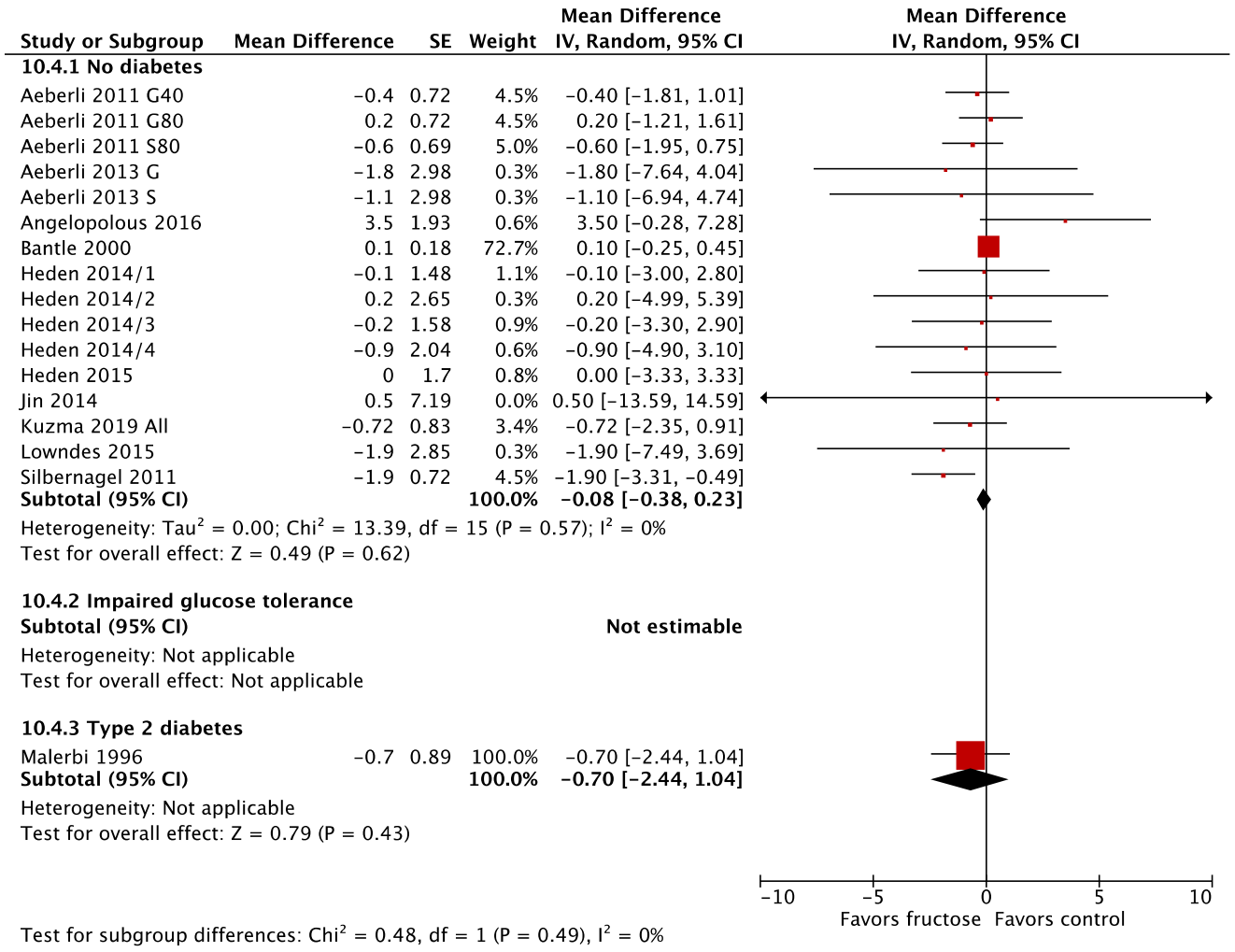
Supplementary figure 23: Subgroup meta-analysis of fasting blood triglycerides following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by dose of sugar. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood triglycerides after fructose consumption and fasting blood triglycerides following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



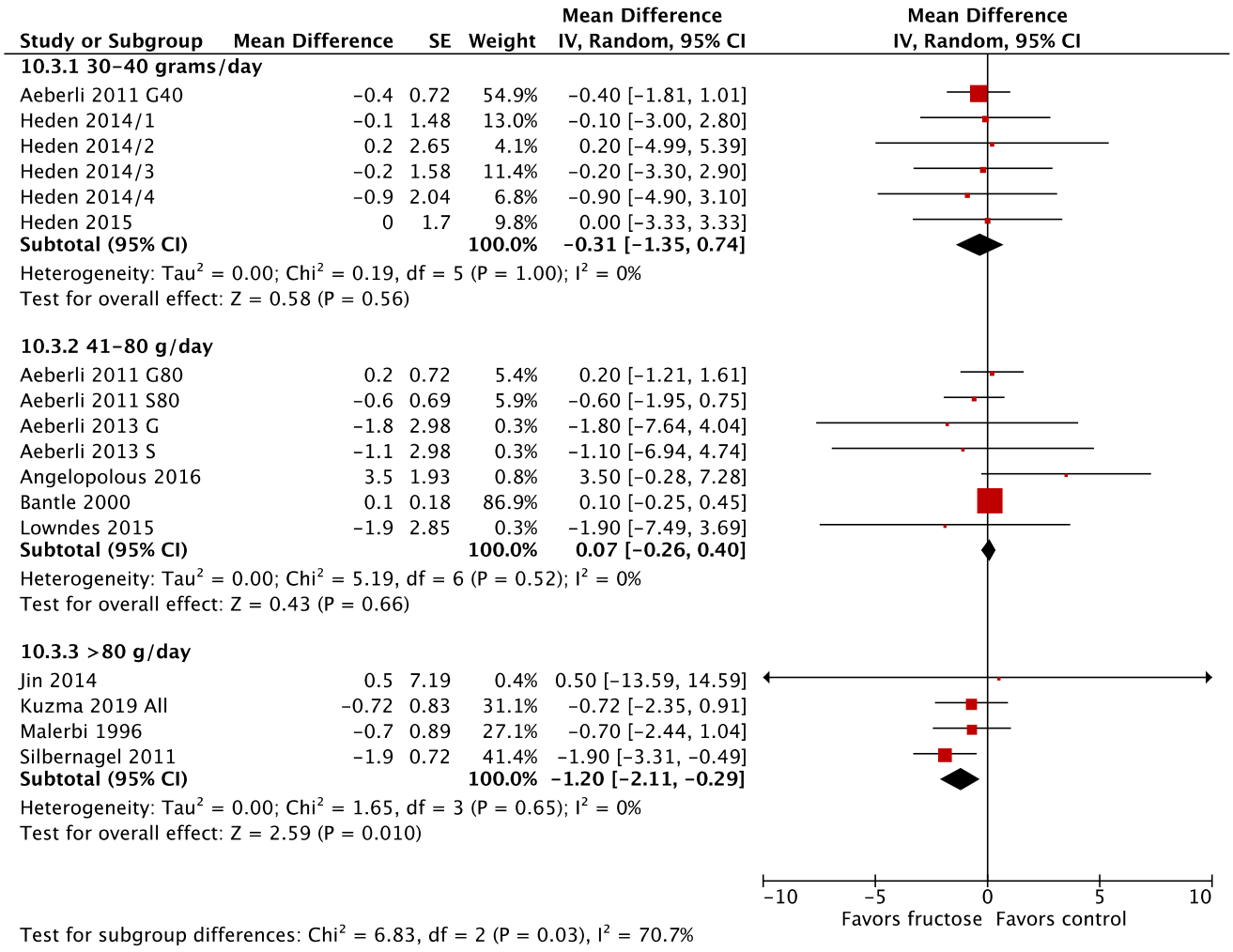
Supplementary figure 24: Subgroup meta-analysis of fasting blood triglycerides following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by diabetes status. Values are mean differences [95% CIs] (expressed as mmol/L) between fasting blood triglycerides after fructose consumption and fasting blood triglycerides following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



Supplementary figure 25: Subgroup meta-analysis of body weight following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by baseline BMI. Values are mean differences [95% CIs] (expressed as kg) between body weight after fructose consumption and body weight following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



Supplementary figure 26: Subgroup meta-analysis of body weight following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by diabetes status. Values are mean differences [95% CIs] (expressed as kg) between body weight after fructose consumption and body weight following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight



Supplementary figure 27: Subgroup meta-analysis of body weight following isoenergetic substitution of glucose or sucrose by fructose in food or beverages by dose of sugar. Values are mean differences [95% CIs] (expressed as kg) between body weight after fructose consumption and body weight following glucose or sucrose consumption. IV, inverse variance; SE, standard error; G40, fructose/glucose 40 g/day; G80, fructose/glucose 80 g/day; G, glucose; S, sucrose; I, impaired glucose tolerance, N, normal glucose tolerance/body weight; O, overweight

Supplementary table 1: meta-regression of studies reporting on fasting blood insulin

Potential covariant		N study arms	Coefficient	Lower bound	Upper bound	p value
Factor covariates						
Control sugar	glucose	13				
	sucrose	3	1.028	-1.543	3.699	0.433
Diabetes status	normoglycaemia	14				
	impaired glucose tolerance	1	-1.874	-4.890	1.142	0.223
	type 2 diabetes	1	-1.024	-5.015	2.967	0.615
Study type	cross-over	13				
	parallel	3	1.905	-1.323	5.133	0.247
Food vs beverage	beverage	12				
	food	4	-0.404	-2.426	1.618	0.695
Gender	males	4				
	both	10	-2.961	-4.109	-1.813	**<0.001**
	females	2	-2.239	-3.622	-0.856	**0.002**
Funding	government	14				
	both	1	-0.677	-4.651	3.297	0.739
	industry	1	4.263	-0.445	8.971	0.076
Blinding	Yes	11				
	Unclear	5	-2.587	-4.187	-0.987	**0.002**
Continuous covariates						
BMI	Range: 22.4 - 34.6	16	-0.061	-0.339	0.217	0.668
Age	Range: 13.5 - 54.2	16	-0.066	-0.125	-0.008	**0.026**
Dose	Range: 35 - 150	16	-0.005	-0.032	0.021	0.689
Year of publication	Range: 1984 - 2019	16	0.089	0.022	0.155	**0.009**

Supplementary table 2: meta-regression of studies reporting on body weight

Potential covariant		N study arms	Coefficient	Lower bound	Upper bound	p value
Factor covariates						
Control sugar	glucose	14				
	sucrose	3	-0.608	-1.703	0.488	0.277
Diabetes status	normoglycaemia	16				
	impaired glucose tolerance	0	N/A	N/A	N/A	N/A
	type 2 diabetes	1	-0.485	-2.345	1.376	0.610
Study type	cross-over	13				
	parallel	4	-1.233	-2.550	0.084	0.066
Food vs beverage	beverage	16				
	food	1	-0.485	-2.345	1.376	0.610
Gender	males	7				
	both	8	0.244	-0.576	1.064	0.560
	females	2	-0.169	-2.730	2.392	0.897
Funding	government	14				
	both	1	-0.416	-2.306	1.474	0.666
	industry	2	2.027	-1.121	5.266	0.203
Blinding	Yes	14				
	Unclear	3	0.628	-0.051	1.307	0.070
Continuous covariates						
BMI	Range: 22.4 - 34.6	17	-0.018	-0.226	0.189	0.862
Age	Range: 13.5 - 54.2	17	0.021	-0.016	0.058	0.270
Dose	Range: 35 - 150	17	-0.017	-0.031	-0.003	**0.016**
Year of publication	Range: 1996 - 2016	17	-0.039	-0.090	0.012	0.131

Supplementary table 2: meta-regression of studies reporting on fasting blood triglycerides

Potential covariant		N study arms	Coefficient	Lower bound	Upper bound	p value
Factor covariates						
Control sugar	glucose	12				
	sucrose	3	-0.096	-0.263	0.071	0.261
Diabetes status	normoglycaemia	13				
	impaired glucose tolerance	1	-0.252	-0.520	0.016	0.066
	type 2 diabetes	1	-0.142	-0.358	0.075	0.199
Study type	cross-over	12				
	parallel	3	-0.001	-0.235	0.233	0.995
Food vs beverage	beverage	11				
	food	4	-0.139	-0.273	-0.005	**0.042**
Gender	males	5				
	both	7	-0.200	-0.322	-0.078	**0.001**
	females	3	-0.165	-0.301	-0.030	**0.017**
Funding	government	13				
	both	1	-0.120	-0.360	0.120	0.328
	industry	1	0.060	-0.240	0.360	0.694
Blinding	Yes	9				
	Unclear	6	-0.031	-0.175	0.112	0.670
Continuous covariates						
BMI	Range: 22.4 - 32.6	15	-0.007	-0.033	0.019	0.609
Age	Range: 17.1 - 54	15	-0.001	-0.006	0.004	0.793
Dose	Range: 35 - 150	15	0.000	-0.003	0.002	0.802
Year of publication	Range: 1984 - 2016	15	0.003	-0.003	0.009	0.314